

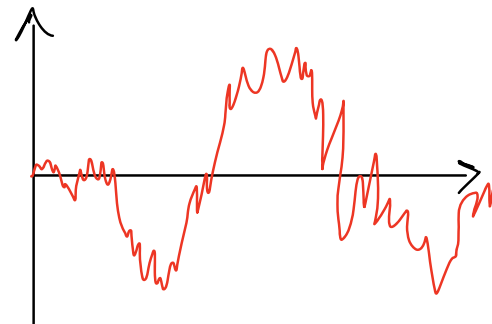
Recitation 3 - Amer AL-Hiyasat

Topics:

- 1) Conditional distributions + constructive definition of the Wiener process
 - 2) General Brownian motions
 - 3) Stochastic Differential Equations
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Recap: Wiener process $W(t)$ is defined by:

- i) Continuous paths
- ii) Stationary, independent increments
- iii) $W(t) \sim \mathcal{N}(0, t)$



An equivalent definition is:

$W(t)$ is a Gaussian process with $\langle W(t) \rangle = 0$ and $\langle W(t_1) W(t_2) \rangle = \min(t_1, t_2)$.

The physics definition of $W(t)$ is that it is the solution to the Langevin eqn:

$$\dot{x} = \eta(t), \quad \text{where } \langle \eta(t) \eta(t') \rangle = \delta(t_1 - t_2)$$

Thus

$$W(t) = \int_0^t ds \, \eta(s)$$

so that we may write

$$'' \frac{dW}{dt} = \eta(t) ''$$

where the quotes emphasize that $W(t)$ is not differentiable in the rigorous sense.

Conditional distributions

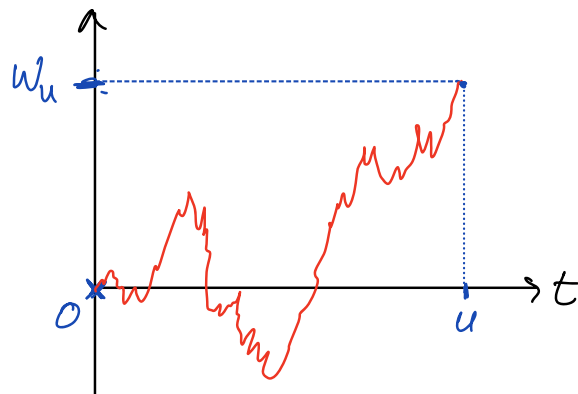
Consider a Wiener Process $W(t)$. Suppose I give you $W(u)$ for some $u > 0$. What is the conditional distribution of $(W(t \leq u) | W(u))$?

The following fact is useful:

Claim: $W_t - (t/u)W_u$ is independent of W_u for $t \leq u$.

PF: Clearly the quantity is Gaussian, so it suffices to show that the covariance is zero.

$$\begin{aligned} \langle (W_t - \frac{t}{u}W_u) W_u \rangle_c &= \min(t, u) - \frac{t}{u} \langle W_u^2 \rangle_c \\ &= t - \frac{t}{u}u = 0. \end{aligned}$$



This makes life easy:

$$\begin{aligned} \langle W_t | W_u \rangle &= \langle W_t - (t/u)W_u | W_u \rangle + \frac{t}{u}W_u \\ &= \langle W_t - \frac{t}{u}W_u \rangle + \frac{t}{u}W_u \\ &= \boxed{\frac{t}{u}W_u} \end{aligned}$$

This makes sense: the mean is just obtained by linearly interpolating b/w the two points. The variance is:

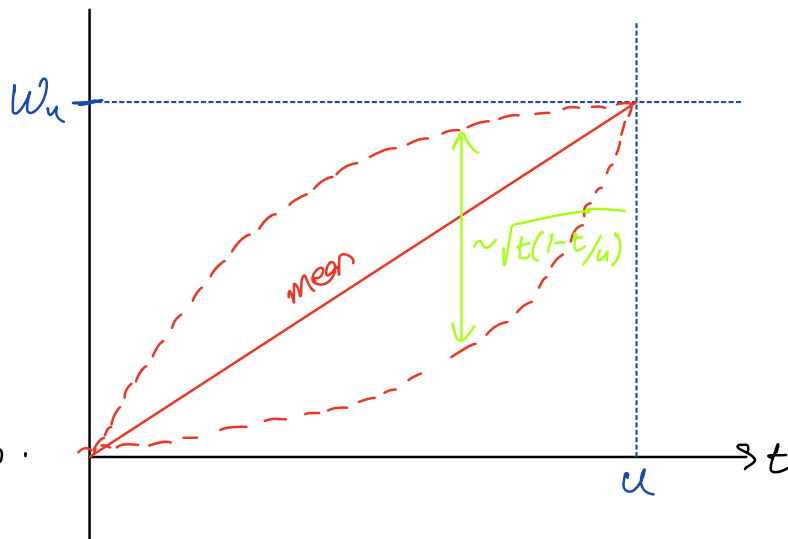
$$\begin{aligned} \langle W_t^2 | W_u \rangle &= \langle (W_t - \langle W_t | W_u \rangle)^2 | W_u \rangle \\ &= \langle (W_t - \frac{t}{u}W_u)^2 | W_u \rangle \\ &\stackrel{\text{by claim}}{=} \langle (W_t - \frac{t}{u}W_u)^2 \rangle \\ &= t - 2\frac{t}{u}\min(t, u) + \frac{t^2}{u^2}u \\ &= \boxed{t(1 - t/u)} \end{aligned}$$

As expected, variance is smallest near t or u , and maximized at $t = u/2$.

Strangely, this is independent of W_u !

As W_u is made larger, the relative width vanishes:

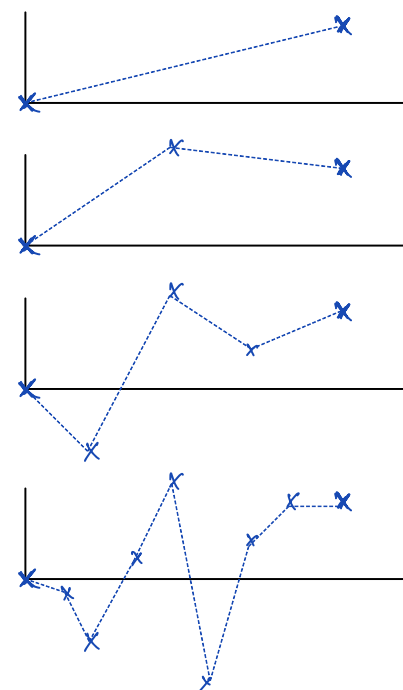
$$\frac{\sqrt{\langle W_t^2 | W_u \rangle_c}}{\langle W_t | W_u \rangle} \leq \frac{u/2}{t/u W_u} \rightarrow 0 \text{ as } W_u \rightarrow \infty.$$



(Maximum variance is always $u/4$)

Note that the result above gives a procedure for sampling a Wiener process on the interval $(0, u)$

- 1) Set $w(0) = 0$.
- 2) Set $w(u)$ by drawing a $\mathcal{N}(0, u)$ number.
- 3) Set $w(u/2)$ by drawing a $\mathcal{N}(\frac{1}{2}w(u), u/4)$
- 4) Repeat for the intervals $(0, u/2)$ and $(u/2, u)$, continue recursively.



The conditional correlation function is, for $s < t < u$

$$\langle w(s)w(t) | w(u) \rangle_c = \frac{s(u-t)}{u}$$

I leave the proof as an exercise.

The Brownian Bridge

In the special case $u=1$ and $w_u = 0$, the resulting conditional distribution is called a Brownian Bridge. This can be uniquely defined as follows

Def: A standard Brownian Bridge is a Gaussian process $\{X(t): t \in [0, 1]\}$ with continuous paths, mean zero, and $\langle X(s)X(t) \rangle_c = s(1-t)$ for $0 \leq s \leq t \leq 1$.

You can verify that if w is a Wiener process, the following give Brownian bridges:

$$X(t) = w(t) - tw(1)$$

$$Y(t) = (1-t)w\left(\frac{t}{1-t}\right)$$

You can interpret this as a Wiener process "pinned" at $w(1)=0$.

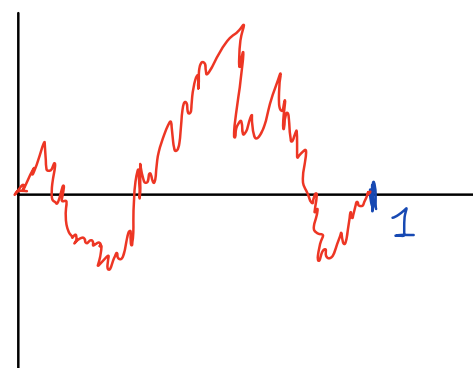
You may verify that a Brownian bridge has the

following Fourier representation:

$$B_t = \sum_{k=1}^{\infty} \frac{Z_k}{k} \frac{\sqrt{2} \sin(k\pi t)}{\pi}$$

where the $\{Z_k\}$ are i.i.d $\mathcal{N}(0, 1)$.

The spectral density then falls as $1/k^2 \Rightarrow$ this is like a simple Gaussian field theory with Hamiltonian $\mathcal{H}[\psi] = \int dx (\nabla \psi)^2$ (don't worry if this is meaningless to you).



General Brownian motions (with drift)

We may generalize the Wiener process by adding a drift and a scale to $W(t)$.

A process $X(t)$ defined as

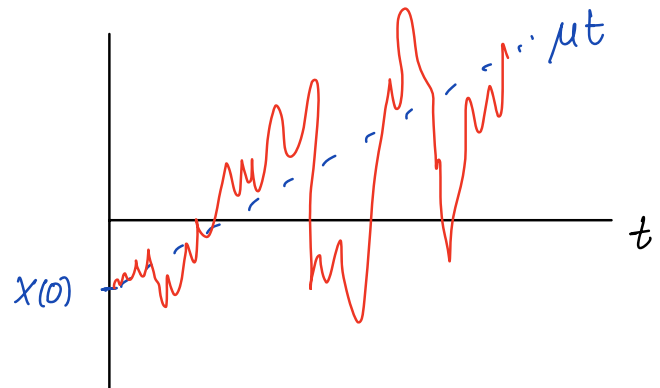
$$X(t) = X(0) + \mu t + \sigma W(t)$$

is called a (μ, σ^2) -Brownian motion

(though this terminology is rarely used in physics).

I bring this up because of an important fact:

Thm: If a stochastic process X has continuous paths and stationary independent increments, then X is a Brownian motion as defined above.



The Gaussianity of increments thus comes "for free". This can be intuited from the central limit theorem: If increments are stationary and independent, then any one increment can be partitioned as the sum of many smaller increments that are i.i.d.

Stochastic differential equations

The solution to a (deterministic) ODE of the form

$$\dot{x} = f(x)$$

is a smooth function $x(t)$ (so long as f is smooth).

Such a solution can everywhere be locally approximated by a linear function:

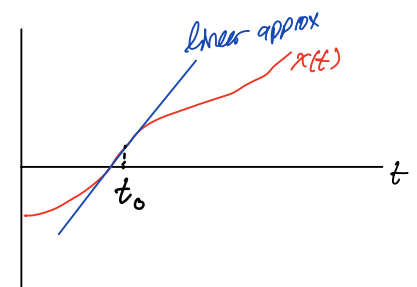
$$x(t) \simeq x(t_0) + f(x(t_0))(t - t_0) \quad \text{as } t \rightarrow t_0$$

A stochastic differential equation

$$\dot{x} = f(x) + \sigma(x) \eta(t)$$

has solutions $x(t)$ which can be locally approximated by a general Brownian motion:

$$x(t) \simeq x(t_0) + f(x(t_0))(t - t_0) + \sigma(x(t_0)) W(t - t_0) \quad \text{as } t \rightarrow t_0$$



Stochastic Integrals

A stochastic differential equation is a shorthand for the corresponding stochastic integral:

$$x(t) = \int_0^t ds f(x(s)) + \int_0^t ds \sigma(x(s)) \eta(s)$$

In math, integrals of the form $\int_{t_0}^t ds \eta(s) h(s)$ are typically denoted

$$\int_{t_0}^t ds \eta(s) h(s) = \int_{t_0}^t dW(s) h(s) \quad (1)$$

i.e. we replace $ds \eta(s) \rightarrow dW(s)$.

The corresponding notation for an SDE is

$$dx = f(x,t) dt + \sigma(x,t) dW$$

We define (1) as a stochastic version of the Riemann-Stieltjes integral:

Partition the interval $[t_0, t]$ into n subintervals,

$$t_0 \leq t_1 \leq \dots \leq t_{n-1} \leq t$$

Pick times τ_i within these subintervals:

$$t_{i-1} \leq \tau_i \leq t_i$$

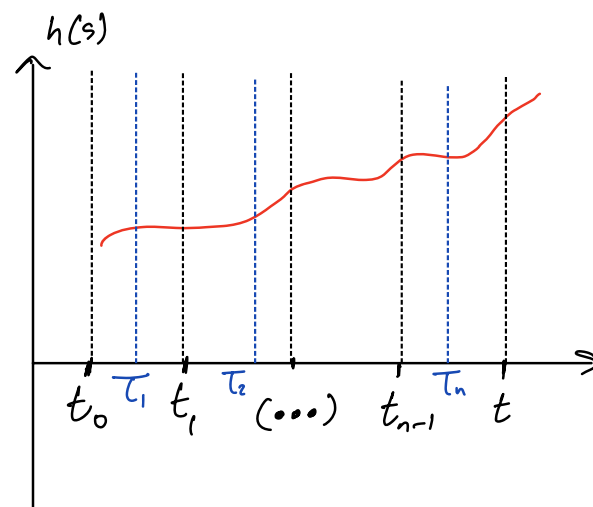
The stochastic integral is defined as the limit $\int_{t_0}^t h(s) dW(s) = \lim_{n \rightarrow \infty} S_n$, where

$$S_n \equiv \sum_{i=1}^n h(\tau_i) [W(t_i) - W(t_{i-1})]$$

where W is a standard Wiener process.

Note that the value of this integral depends on the choice of τ_i . For example, consider $h(s) = W(s)$. Then:

$$\begin{aligned} \langle S_n \rangle &= \sum_{i=1}^n \langle W(\tau_i) [W(t_i) - W(t_{i-1})] \rangle \\ &= \sum_{i=1}^n [\min(\tau_i, t_i) - \min(\tau_i, t_{i-1})] \\ &= \sum_{i=1}^n (\tau_i - t_{i-1}) \end{aligned}$$



If we use an " α -discretization" as mentioned in lecture:

$$t_i \equiv \alpha t_i + (1-\alpha) t_{i-1}, \quad 0 \leq \alpha \leq 1,$$

Then

$$\langle S_n \rangle = \sum_{i=1}^n \alpha (t_i - t_{i-1}) = \alpha (t - t_0)$$

This ranges anywhere between 0 and $t - t_0$.

In the Itô prescription, we make the choice

$$\alpha = 0$$

In the Stratonovich prescription, we use

$$\alpha = 1/2.$$

For arbitrary $h(s)$, there is no general correspondence between the two integrals, but in the important special case where $h(s) = h(x(s))$ and $x(s)$ is the solution to an SDE, there is in fact a general formula relating the two prescriptions. We will come back to this in a future recitation.